MILITARY SPECIFICATION

CABLE ASSEMBLY MOLDED CONNECTOR; GENERAL PURPOSE, ELECTRONIC EQUIPMENT (GROUNDING CONNECTOR MOLDED)

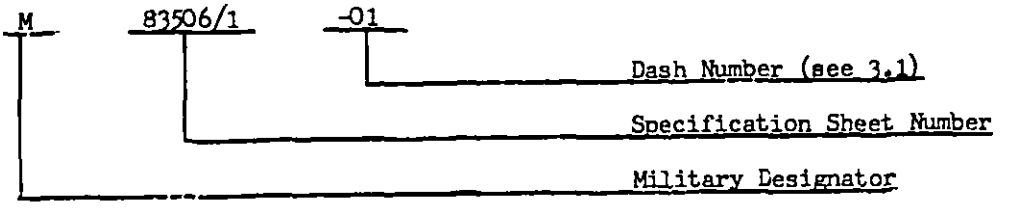
"This specification is approved for use by The Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by all Departments Agencies of the Department of Defense."

1. SCOPE.

1.1 This specification covers cable assemblies having molded grounding connectors (plugs and cabled outlets) attached.

1.2 Terms Electrical: Specific electrical terms referred to in this document are defined in WD1-1974 and WD5-1974, Revision 2.

1.3 Military Part Number. The Military Part Number shall consist of the letter M, the basic number of the specification sheet, and an assigned dash number (see 3.1), as shown in the following:



2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-631	-	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-C-3432	-	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-I-3930	-	Insulating and Jacketing Compounds, Electrical (for Cables, Cords and Wires)
MIL-C-45662	-	Calibration System requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering and Technical Support Division (AFLC/LOIE), Wright-Patterson AFB, Dayton OH 45433, by using the self-addressed Standardization Document Improvement Proposal DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1285 - Marking of Electrical and Electronic Parts
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Electrical Manufacturers Association (NEMA) Publications:

WD1 - General Purpose Wiring Devices.
WD5 - Specific Purpose Wiring Devices.

(Application for copies should be addressed to National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of the specification and the specification sheets, the latter shall govern.

3.2 First article. When specified (see 6.3), the contractor shall furnish sample units for first article inspection and approval (see 4.4).

3.3 Cable assemblies.

3.3.1 Cable assemblies shall be furnished with a nonremovable grounding connector molded to the cable jacket.

3.3.2 Cable. The cable shall be a 3 conductor flexible cord conforming to type CO-03NGF(3/18)0340 of MIL-C-3432 with the leads color-coded black, white, and green. The cable shall be 8 feet ± 3 inches long. The color of the jacket of the cable shall be black.

3.4 Connectors.

3.4.1 The connectors shall be of the type molded to the cable jacket. Configuration shall be as specified (see 3.1).

3.4.2 Materials. All current carrying and grounding parts shall be copper-base alloy. Insulating material shall be per MIL-I-631 or MIL-I-3930.

3.4.3 Wiring. Wiring shall be as specified (see 3.1).

3.5 Finish. Connectors shall have surfaces which are not warped, cracked, chipped, or blistered either initially or as a result of the tests specified herein.

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3.6 Performance.

3.6.1 Gripping (straight blade outlets only). When connectors are tested as specified in 4.6.2, the test blade or pin shall not be displaced from the face of the connector under test by more than .06 inch (1.52 mm).

3.6.2 Continuity. Each conductor to contact in the assembly shall be continuous when tested as specified in 4.6.3.

3.6.3 Current overload. When tested as specified in 4.6.4 assemblies shall meet the requirements of 3.6.2.

3.6.4 Dielectric withstanding voltage. When assemblies are tested as specified in 4.6.5 there shall be no damage, arcing or breakdown.

3.6.5 Insulation resistance. When assemblies are tested as specified in 4.6.6 insulation resistance of mutually insulated parts shall be not less than 100 megohms.

3.6.6 Heat resistance. After assemblies are tested as specified in 4.6.7 there shall be no evidence of mechanical or electrical failure, flow or critical softening of sealing compounds, or softening or distortion of parts, assemblies shall then conform to 3.6.5 and 3.6.4.

3.6.7 Mechanical shock. When assemblies are tested as specified in 4.6.8 there shall be no evidence of cracking, breaking or loosening of parts, assemblies shall conform to 3.6.2 and 3.6.5.

3.6.8 Compression (connectors). When connectors are tested as specified in 4.6.9 there shall be no evidence of chipping, cracking, breaking or distortion.

3.6.9 Cable retention forces. When assemblies are tested as specified in 4.6.10 and the force exerted, shall not cause movement of the cable jacket that will result in visible exposure of the insulated conduction (see 3.1), assemblies shall conform to 3.6.2 and 3.6.4.

3.6.10 Contact retention forces (plugs). When plugs are tested as specified in 4.6.11 the contacts shall not loosen, change in shape or position, or be otherwise damaged. Assemblies shall then be tested for conformance to 3.6.2 and 3.6.4.

3.7 Marking. Unless otherwise specified (see 3.1) each cable assembly shall have positioned in the middle of it a marker band of compatible material marked in accordance with MIL-STD-1285 and the following.

a. Military Part Number

b. Manufacturer's Code, Year of Manufacture

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.1.2 Test connector. A test connector of the type specified as the mating connector (see 3.1) shall be used in testing sample connectors. Approval of the test connector for use in the tests shall not constitute acceptance of the connector part number represented by the test connector. In the event of failure of the test connector, a new one shall be substituted and the tests continued. Rejection of a test connector will not constitute a failure of the connector under test.

4.2 Classification of tests. The inspection and testing of the cable assemblies shall be classified as follows:

a. First Article Inspection (see 4.4)

b. Quality Conformance Inspection (see 4.5 and 4.1).

4.3 Inspection conditions and methods.

4.3.1 Conditioning during inspection. Neither pin, blade, nor socket contacts of connectors submitted for examinations or tests shall be adjusted, lubricated or otherwise conditioned before, during, or after any examination or test.

4.3.2 Inspection conditions. Unless otherwise specified herein, inspections shall be performed in accordance with the test condition specified in WD1.

4.4 First Article Inspection. First Article Tests shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 First Article Inspection. Six assemblies shall be subjected to First Article Tests specified in Table I in the order shown.

TABLE I FIRST ARTICLE INSPECTION

Examination or Test	Requirement Paragraph	Method Paragraph
Visual & Mechanical	3.1, 3.3, 3.4, 3.5	4.6.1
Marking	3.7	
Gripping	3.6.1	4.6.2
Continuity	3.6.2	4.6.3
Current Overload	3.6.3	4.6.4
Dielectric Withstanding Voltage	3.6.4	4.6.5
Insulation Resistance	3.6.5	4.6.6
Heat Resistance	3.6.6	4.6.7
Mechanical Shock	3.6.7	4.6.8
Compression	3.6.8	4.6.9
Cable Retention Forces	3.6.9	4.6.10
Contact Retention Forces	3.6.10	4.6.11

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4.5 Quality Conformance Inspection.

4.5.1 Inspection of Product for Delivery. Inspection of product for delivery shall consit of a Group A inspection (see 4.5.1.2).

4.5.1.1 Inspection Lot. An inspection lot shall consist of cable assemblies from all assembly lines, produced essentially under the same conditions and offered for inspection during a single work month.

4.5.1.2 Group A Inspection. Group A inspection shall consist of the examination and test specified in Table II, in the order shown.

4.5.1.2.1 Sampling Plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be one percent defective.

TABLE II GROUP A INSPECTION

Examination or Test	Requirement Paragraph	Method Paragraph
Visual & Mechanical	3.1, 3.3, 3.4, 3.5	4.6.1
Marking	3.7	
Cable Retention Forces	3.6.9	4.6.10
Contact Retention Forces	3.6.10	4.6.11

4.6 Methods of examination and tests.

4.6.1 Visual and mechanical examination. (See 3.1, 3.3, 3.4 and 3.5) Devices shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.2 Gripping (straight blade outlets only). (See 3.6.1) The tests required in 4.6.2.1 and 4.6.2.2 shall be performed using conditioning tools and pull out tools conforming to figures 3-1, 3-2, 3-3, and 3-4 of WD1 Part 3.

4.6.2.1 Power contacts. Power contacts shall be tested in accordance with WD1-3.03 paragraph A.

4.6.2.2 Grounding contacts. Grounding contacts shall be tested in accordance with WD1-3.03 paragraph B.

4.6.3 Continuity. (See 3.6.2) Each conductor of each assembly shall be tested for continuity and correct connections between its terminations, using a potential of not more than 10 volts. A suitable indicator such as an incandescent light bulb shall be placed in the circuit to determine conformance to 3.5.2.

4.6.4 Current overload. (See 3.6.3) Mating connectors shall be wired with suitable conductors, assemblies shall then be tested per WD1-3.04; except test is applicable to the complete assembly.

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4.6.5 Dielectric withstanding voltage. (See 3.6.4) An alternating current root mean square voltage of commercial-line-frequency, equal to 1,000 volts plus twice the rated voltage, shall be applied between mutually insulated terminals and conductors. The test voltage shall be increased from zero at a rate not to exceed 500 volts per second until the specified potential is reached and shall be maintained at the value for one minute. The test voltage may be applied instantaneously for production test at the option of the manufacturer. The assembly shall be examined to determine conformance to 3.6.4.

4.6.6 Insulation resistance. (See 3.6.5) The insulation resistance shall be measured between mutually insulated terminals and conductors. Measurements shall be made by means of a megohm bridge or other approved method at a direct current potential of a least 500 volts to determine conformance to 3.6.5.

4.6.7 Heat resistance. (See 3.6.6) Assemblies shall be placed in an oven maintained at 85° ± 2°C (185° ± 4°F) for a period of at least two hours. After removal from the oven, the connectors shall be examined for conformance to 3.6.6.

4.6.8 Mechanical shock. (See 3.6.7) Assemblies shall be tested using test machine illustrated in Fig 3-10 of WD1 when hanging freely, the plug takes the position (1). A moving member shall lift the connector assembly to the position (2) and then release so that it falls freely and strikes impact board B. This constitutes one cycle of the test. Each sample shall be subjected to the above test for 1,300 cycles, then tested for dielectric withstanding voltage, insulation resistance, and continuity as specified in 3.6.4, 3.6.5 and 3.6.2. The samples shall be considered acceptable if each sample completes not less than 500 cycles and the average number of cycles completed for all plugs tested is not less than 1,000. Samples shall be inspected every 50 cycles.

4.6.9 Compression (connectors). (See 3.6.8) The connector shall be placed between two parallel, hard wood slabs, each at least .50 inch thick, with the major axis of the device parallel with the flat surfaces of the slabs. A compression force shall be applied to the connector at right angles to the major axis and shall be increased to 75 pounds and maintained at that value for one minute. The connector shall be examined after the test for conformance to 3.6.8.

4.6.10 Cable retention forces. (See 3.6.9) With the connector held securely in place, a mechanical force shall be applied to the cable in a direction tending to withdraw the cable from the plug. The force shall be increased to 30 pounds minimum and held at that value for one minute. Movement of the jacket shall not exceed that specified.

4.6.11 Contact retention forces (plugs). (See 3.6.10) Plugs shall be held securely in place and a force of 20 pounds minimum shall be applied to each contact in a longitudinal direction away from the contact and shall be held at that value for at least one minute.

5. PACKAGING.

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55442.

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6. NOTES.

6.1 Intended use. The connector cable assemblies covered by this specification are intended for use with test equipment having a capability of operating from 50 to 400 Hz, 115 and/or 230 VAC power sources. Equipment current requirements are limited to 10 amperes at 250 VAC and 15 amperes at 115 VAC.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Complete part number (see 1.3).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Design activity. The activity responsible for this specification is Air Force Code 85, AFLC/LOIEP Wright-Patterson AFB OH 45433.

Custodian:
Air Force - 85

Review activities:
Air Force - 99

Preparing activity:
Air Force - 85

(Project Number 5995-F097)

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-83506 (USAF) Cable Assembly Molded Connector, FSC 5995

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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